

# WRDS RESEARCH PROJECT

Financial Performance Comparison: NVIDIA vs. Intel

|                    |                                                      |
|--------------------|------------------------------------------------------|
| Course             | MSBA-235-01, Spring 2026                             |
| Professor          | Chong Huang                                          |
| Team Members       | Nhi Nguyen Ngoc Mai (Lucky)<br>Ying Tzu Chen (Wendy) |
| Data Source        | Wharton Research Data Services (WRDS) – Compustat    |
| Companies Analyzed | NVIDIA Corporation (NVDA)   Intel Corporation (INTC) |
| Year               | 2026                                                 |

This report presents a research project on the Wharton Research Data Services (WRDS) platform, developed for MSBA-235-01. It introduces WRDS as a research infrastructure, reviews a peer-reviewed paper on WRDS accounting databases, and applies WRDS (Compustat) data to a case study comparing the financial performance of NVIDIA and Intel across revenue, profitability, and asset efficiency.

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# 1. Introduction to WRDS

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## 1.1 What is WRDS?

- **Full name:** Wharton Research Data Services (WRDS)
- **Developed by:** The Wharton School, University of Pennsylvania
- **Format:** Web-based data platform

### Purpose

- Provides financial, economic, and market data
- Supports academic and business research

### Target Users

- MBA / MSBA / PhD students
- Faculty researchers
- Financial analysts
- Institutional investors
- Consulting firms

## 2. Why WRDS Is Important

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- Delivers clear and standardized accounting information
- Maintains historical consistency among companies and over time
- Frequently referenced in academic studies
- Supports data-driven decision-making
- Allows integration with statistical tools (SAS, Python, R)

WRDS enables researchers to combine multiple datasets in a consistent format, thereby facilitating cross-company comparisons and long-term analyses (SSRN, 2017).

*WRDS is not just a database – it is a research infrastructure.*

### 3. Peer-Reviewed Paper Review

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**Article title:** International Accounting Databases on WRDS: Comparative Analysis

**Reference:** Dai, R. (2012). International accounting databases on WRDS: Comparative analysis. Available at SSRN 2938675, <http://dx.doi.org/10.2139/ssrn.2938675>

**Methodology:** Secondary data from multiple accounting databases available on WRDS:

- Compustat Global
- FactSet Fundamentals
- Bureau van Dijk (BvD / Osiris)

### 4. How the Paper Contributes Data to WRDS

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#### Data Sources

- Compustat → firm financial statements (revenue, net income)
- CRSP → stock prices and returns

#### How the Data Is Used

- Combine financial and market data into a unified dataset
- Analyze firm performance and test financial relationships
- Enable consistent, cross-company and time-series comparison

#### Research Contribution

- Demonstrates integration of standardized WRDS datasets
- Applies real-world financial variables in empirical analysis
- Enhances reliability and credibility of financial research
- Does not create new data, but maximizes the value of existing datasets

## 5. Main Features of WRDS

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| Key Databases in WRDS                                                                                                              | Core Features                                                                                                   | Strengths of WRDS                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Compustat (Capital IQ):<br>Company financial data<br><br>CRSP: Stock prices & returns<br><br>(Optional) IBES: Analyst<br>forecasts | Search companies<br>(name/ticker)<br>Select financial variables<br>Filter by time period<br>Export to Excel/CSV | Standardized datasets<br>High accuracy & consistency<br>Academic-grade data quality<br><br>Ideal for: financial analysis,<br>valuation, market research |

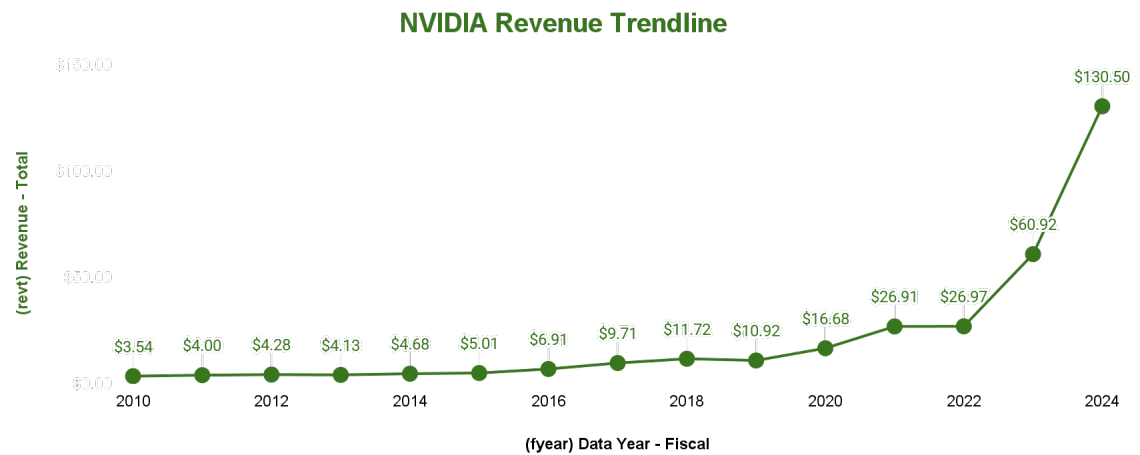
## 6. Case Study Overview: NVIDIA vs. Intel

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| Companies       | Objective                                                     | Data Source         |
|-----------------|---------------------------------------------------------------|---------------------|
| NVIDIA<br>Intel | Analyze financial performance<br>Support investment decisions | Compustat<br>(WRDS) |

# 7. NVIDIA Financial Analysis

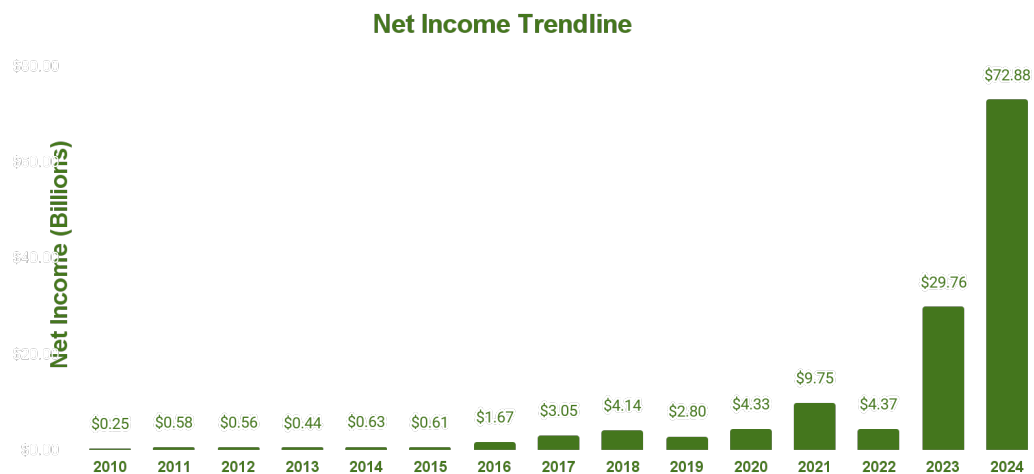
## 7.1 Revenue Analysis



| Period    | Trend                                                             | Driver                                                   |
|-----------|-------------------------------------------------------------------|----------------------------------------------------------|
| 2010–2015 | Slow and stable growth (\$3.5B → \$5B)                            | Focus on gaming GPUs and the traditional graphics market |
| 2016–2020 | Strong upward trend (\$6.9B → \$16.7B), minor fluctuation in 2019 | Expansion into data centers, AI, and machine learning    |
| 2021–2024 | Rapid and explosive growth (\$60.9B → \$130.5B)                   | Driven by the AI boom and high demand for GPU chips      |

NVIDIA’s revenue increases sharply, especially in recent years, indicating strong market demand driven by AI technologies.

## 7.2 Net Income (Loss) Analysis

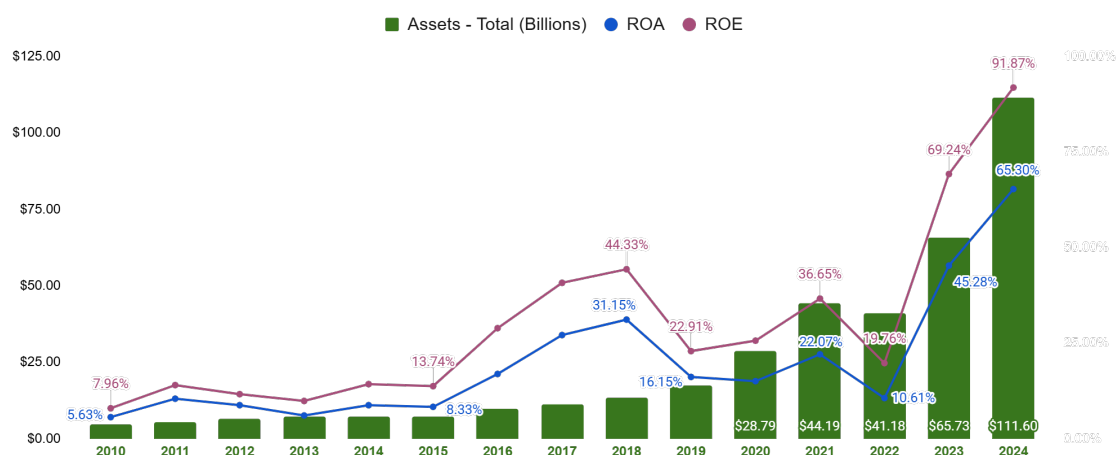


Net income rises exponentially.

| Period    | Observation                                                                                                                                                                                                                                                       |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2018–2022 | Fluctuations, with a slight dip in 2022 – driven by market cycles, supply chain adjustments, and inventory costs. The 2022 decline was largely due to a 43% drop in Gaming revenue in some quarters. COVID-19 impact was minor, and did not result in a net loss. |
| 2023–2024 | Explosive growth (\$29.8B → \$72.9B), driven by both revenue growth and higher profit margins.                                                                                                                                                                    |

Although labeled as net income (loss), the values are positive and show a sharp increase, reflecting strong profitability and high margins.

## 7.3 Assets & Efficiency Analysis

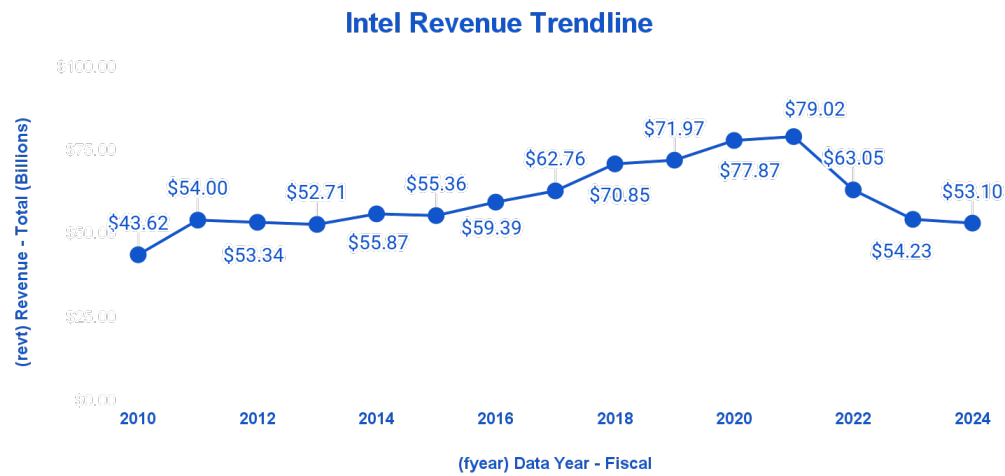


|           | Details                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------|
| Highlight | Strong and increasing ROA and ROE; efficient profit generation with relatively lower assets              |
| Lowlight  | Asset base is smaller compared to competitors                                                            |
| Reason    | Asset-light business model (outsourced manufacturing); high demand for AI products drives strong margins |

**Return on Assets (ROA)** measures how efficiently a company uses its assets to generate profit. **Return on Equity (ROE)** measures how efficiently management uses investor money.

# 8. Intel Financial Analysis

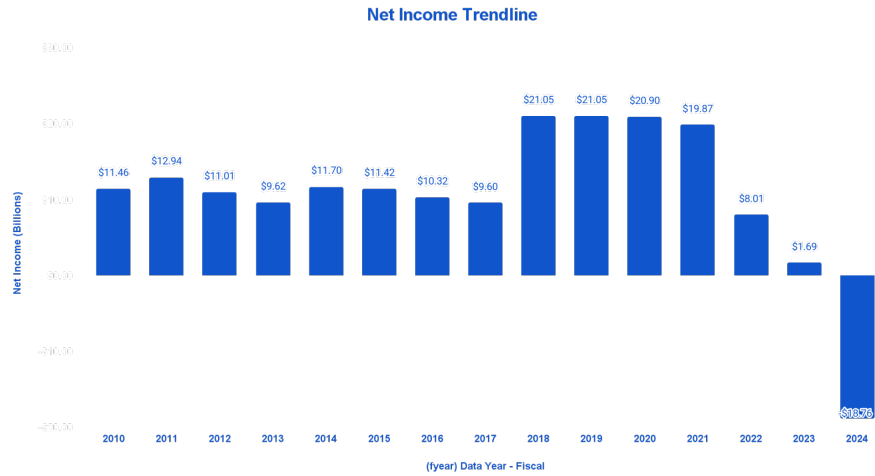
## 8.1 Revenue Analysis



| Period    | Trend                                                    | Driver                                                       |
|-----------|----------------------------------------------------------|--------------------------------------------------------------|
| 2010–2014 | Mature stability: revenue stable (~\$43B–\$55B)          | Dependence on the PC CPU market                              |
| 2015–2019 | Moderate growth, incremental rather than exponential     | Supported by data center processors                          |
| 2020–2024 | Volatility & decline: revenue declined (~\$79B → ~\$54B) | Competition intensified; limited exposure to the AI GPU boom |

Intel's challenge is transitioning from legacy PC dominance to AI and advanced compute markets.

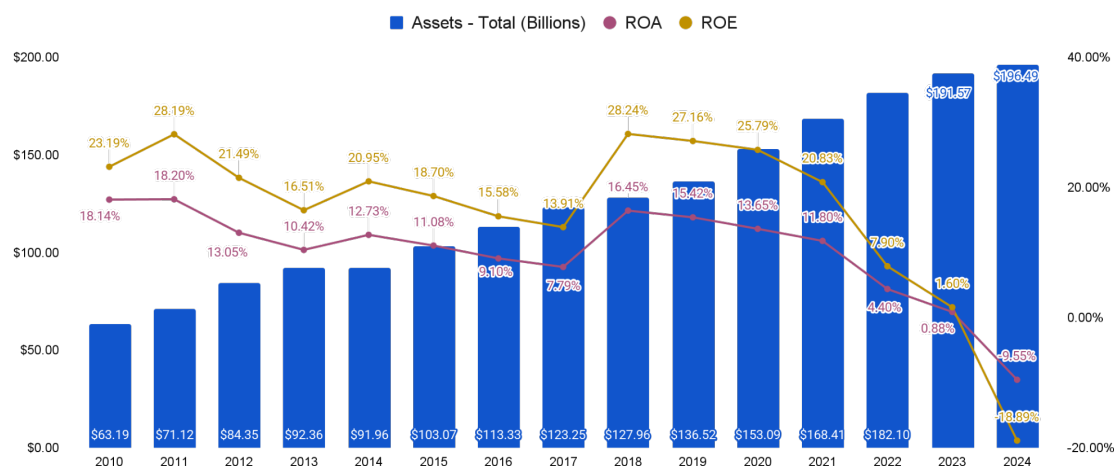
## 8.2 Net Income (Loss) Analysis



| Period    | Observation                                                                                                                                                    |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2020–2023 | Volatility & decline – high volatility and sharp profitability drops due to economic headwinds, supply issues, weaker PC demand, and high capital expenditure. |
| 2023–2024 | Weak recovery – no strong rebound in net income compared to peers like NVIDIA; profitability remains under pressure.                                           |

Intel's earnings are not driven by margin expansion, indicating structural challenges in improving profitability.

## 8.3 Assets & Efficiency Analysis



|           | Details                                                                                                                                                                                                 |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Highlight | Strong asset growth: total assets steadily increased (~\$63B → ~\$196B), showing long-term expansion. Peak ROE/ROA in 2018–2019 reflected high efficiency.                                              |
| Lowlight  | Sharp profitability decline post-2020; ROE and ROA turned negative by 2024; asset growth not generating returns.                                                                                        |
| Reason    | Heavy capital investments (fabs and manufacturing transformation) increased the asset base; weak PC demand (post-COVID) and rising competition (e.g., NVIDIA); margin pressure and restructuring costs. |

## 9. Comparison: NVIDIA vs. Intel

(Data via WRDS – Compustat)

| Aspect (via WRDS – Compustat) | NVIDIA                                        | Intel                                 |
|-------------------------------|-----------------------------------------------|---------------------------------------|
| Revenue (REVT)                | Rapid growth, especially after 2022 (AI boom) | Stable, then decline after 2021       |
| Net Income (NI)               | Strong increase, peak in 2023–2024            | Fluctuating, declining profitability  |
| Profitability (ROA, ROE)      | High and improving efficiency                 | Declining efficiency over time        |
| Asset Growth (AT)             | Fast growth, supports expansion               | Large growth but capital-intensive    |
| Equity (CEQ)                  | Strong and consistently increasing            | Positive but slower growth            |
| Growth Driver (Insight)       | AI, GPUs, data centers                        | CPUs, traditional PC market           |
| Performance Evaluation        | High-growth, high-margin company              | Stable but under performance pressure |

**Using WRDS standardized accounting data:**

- **NVIDIA:** Growth driven by high-margin AI demand
- **Intel:** Capital-intensive and slower growth model

**WRDS enables:**

- Long-term trend tracking
- Cross-firm comparability
- Ratio computation from raw financial data
- Research-grade financial modeling

# 10. Pros and Cons of Using WRDS

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| PROS                                                                         | CONS                                                                                |
|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| High-quality data: reliable, accurate, and standardized datasets             | Access restriction: requires institutional subscription                             |
| Comprehensive coverage: access to multiple databases (Compustat, CRSP, etc.) | Complex interface: not beginner-friendly for first-time users                       |
| Time efficiency: easy data extraction and export (Excel/CSV)                 | Limited real-time data: mostly historical datasets                                  |
| Academic credibility: widely used in research and publications               | Data dependency: users rely heavily on available variables (limited customization)  |
| Supports analysis: enables financial modeling, valuation, and trend analysis | Learning curve: requires understanding of financial variables (REVT, NI, ROA, etc.) |

WRDS is a powerful and reliable research tool, but requires access and technical knowledge to use effectively.

# 11. Conclusion

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## What WRDS Provides

- Evidence-based investment decision support
- Long-term financial performance analysis
- Cross-company comparability
- Support for academic & professional research

## What We Demonstrated (via Compustat)

- Extraction of raw financial data
- Custom financial ratio calculations
- Strategic business insights

*WRDS transforms raw accounting data into strategic business insight.*

## 12. Reference

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Dai, R. (2012). International accounting databases on WRDS: Comparative analysis. Available at SSRN 2938675.  
<http://dx.doi.org/10.2139/ssrn.2938675>

SSRN. (2017). [Cited regarding standardized, cross-company financial research formats enabled by WRDS.]

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